

Input Data Manipulation: Census Data, GEOID, and Shapefile Integrity

R.2 — Adversarial Robustness Track

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Abstract

Algorithmic redistricting depends on three data sources: the Census P.L. 94-171 redistricting file, TIGER/Line shapefiles, and optionally American Community Survey 5-year tables. A hostile actor who can modify any of these sources before the algorithm processes them can potentially steer partisan outcomes without touching algorithm code or parameters.

We analyse the three attack surfaces—population count manipulation, shapefile boundary modification, and ACS demographic table alteration— and show that (1) even a $\pm 1\%$ per-tract population manipulation produces at most $\Delta D \leq 0.4$ expected Democratic seats nationally, placing the effect within the B.7 seed-variance noise floor; (2) all three attack surfaces are protected by BISECT’s three-level SHA-256 audit chain: Census download hash, adjacency construction hash, and final plan hash; and (3) any manipulation that exceeds the noise floor requires modifications large enough to produce detectable hash mismatches.

A disclosed limitation: BISECT uses TLS certificate validation via the system trust store but does not implement certificate pinning. A sophisticated man-in-the-middle attacker with TLS spoofing capability could intercept Census downloads without changing the algorithm’s behavior. We recommend supplementary out-of-band hash verification for litigation-grade deployments.

The three-level audit chain—implemented via `bisect fetch` and `bisect label-verify`—provides a computationally infeasible barrier to undetected manipulation: SHA-256 collision resistance requires 2^{80} operations under the birthday bound (FIPS PUB 180-4 [National Institute of Standards and Technology, 2015]).

Keywords

Input data integrity, SHA-256, Census P.L. 94-171, TIGER/Line shapefiles, ACS tables, audit chain, GEOID, TLS verification, bisect fetch, adversarial robustness

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1. Introduction

Algorithmic redistricting transfers the locus of potential manipulation from the map-drawing stage (where humans directly choose boundaries) to the data preparation stage (where the algorithm’s inputs are assembled). An adversary who cannot influence the algorithm’s parameters or code may still be able to achieve partisan outcomes by manipulating the population counts, geographic boundaries, or demographic tables that the algorithm uses as inputs.

This paper examines input data manipulation as a gaming vector (V2 in the R.0 taxonomy [Della-Libera, 2026d]). We ask: given full access to the data pipeline between the Census Bureau and the redistricting algorithm, what is the maximum partisan shift an adversary can achieve without triggering the SHA-256 audit chain?

The question has both empirical and cryptographic components. The empirical component asks: how much does a given magnitude of data manipulation shift partisan outcomes? The cryptographic component asks: at what manipulation magnitude does the SHA-256 audit chain provide a computationally infeasible barrier to undetected manipulation?

1.1. Data Sources

The BISECT algorithm ingests three classes of data:

1. **Census P.L. 94-171 Redistricting Data:** Population counts for each census tract (GEOID), disaggregated by race and Hispanic origin for VRA compliance. Released every decade by the Census Bureau within 12 months of census day.
2. **TIGER/Line Shapefiles:** Geographic boundaries for each census tract. Released annually by the Census Bureau’s Geography Division. The redistricting-year vintage (e.g., 2020) is the authoritative source.
3. **ACS 5-Year Tables:** American Community Survey estimates of demographic composition, used for VRA boost weight computation. Annually released; BISECT uses the vintage ending in the census year (e.g., 2016–2020 for the 2020 cycle).

All three are public-domain and available directly from the Census Bureau. The key security property is that they are also independently verifiable: any party can download them directly and compare against the algorithm’s cached copies.

1.2. Road Map

Section 2 describes the three attack surfaces. Sections 3–5 analyse each attack in detail. Section 6 describes the three-level SHA-256 audit chain. Section 7 discloses the TLS limitation. Section 8 quantifies the combined partisan effect. Section 9 concludes.

2. Attack Surfaces

Three attack surfaces correspond to the three data sources. Table 1 summarises the attacks, their magnitude, and their detectability under the BISECT audit chain.

Table 1: Input data manipulation attack surfaces. Detection probability under BISECT audit chain. ΔD : estimated maximum Democratic seat shift, national.

Attack	Data source	ΔD	Detection	Method
Population inflation/deflation	P.L. 94-171	≤ 0.4	≈ 1.00	Download hash
Shapefile boundary modification	TIGER/Line	≤ 0.4	≈ 0.98	Adjacency hash
ACS demographic alteration	ACS 5-year	≤ 0.1	≈ 1.00	Download hash
Combined (empirical, saturated)	All three	≤ 0.4	≈ 0.98	

All three attacks require the adversary to control the data pipeline between the Census Bureau’s servers and the BISECT computation. This typically means the adversary must either (a) compromise the redistricting authority’s internal data storage, (b) intercept network traffic during the download phase (the TLS limitation discussed in Section 7), or (c) substitute files after download but before the hash is computed.

Option (c) is the most feasible in practice but is prevented by computing the SHA-256 hash immediately upon download and storing it in the manifest. Options (a) and (b) require infrastructure-level compromise that is beyond the scope of redistricting-specific gaming.

3. Attack Surface 1: Census P.L. 94-171 Population Counts

3.1. The Data Format

The Census P.L. 94-171 redistricting data file provides population counts for every census geographic unit from the state level to the block level. For BISECT, the relevant unit is the

census tract (GEOID prefix ending in “00”, indicating a 2020 census tract). Each tract record contains:

- Total population (P1_001N)
- Population by race (P1_002N through P1_009N)
- Hispanic or Latino origin (P2_002N)
- Voting-age population by race (P3, P4 tables)

The algorithm uses P1_001N (total population) for the equal-population constraint and the P3/P4 tables for VRA boost weight computation.

3.2. The Attack

A hostile actor who controls the P.L. 94-171 file could systematically:

1. Inflate population counts in tracts that vote heavily Republican (rural areas, outer suburbs).
2. Deflate population counts in tracts that vote heavily Democratic (urban cores, college towns).

This distortion biases the equal-population constraint: the algorithm will split tracts into districts that are “equal” in the manipulated population counts but systematically different in true population. The practical effect is that Republican-heavy tracts become overrepresented in the sense that the algorithm assigns fewer such tracts per district than warranted by their true population.

3.3. Effect Magnitude

The key constraint on this attack is averaging. A $\pm 1\%$ manipulation per tract produces a distortion of $\pm 0.01 \times p_i$ people in tract i (where p_i is the true population). For an average-sized congressional district ($\sim 760,000$ people, drawn from ~ 18 tracts in a 14-district state):

$$\text{District-level distortion} = \pm 1\% \times \frac{p_{\text{state}}}{k} \approx \pm 7,600 \text{ people.}$$

To flip one competitive district from Democratic to Republican using population manipulation alone, the adversary must bias the boundaries enough to systematically include Republican-heavy tracts at the expense of Democratic-heavy tracts. Given that competitive districts are typically within 2–3% of the partisan threshold (D -vote share $\in [47\%, 53\%]$), and given that the algorithm adds $\pm 7,600$ people per district through manipulation, the expected vote-share change in a competitive district is approximately:

$$\Delta D\text{-share per district} \approx \pm 1\% \times R = \pm 0.3\%,$$

where R is the ratio of partisan correlation with the manipulation to the district size. This is far below the 3–6% margin of a competitive district.

Nationally, summing across all competitive districts and applying the saturation argument from R.0 [Della-Libera, 2026d]: $\Delta D \leq 0.4$ seats.

3.4. Detection

`bisect fetch` computes:

`SHA256(pl94171_state_XX_2020.pl) -> stored in manifest`

and compares against the Census Bureau’s published checksum at:

`https://www2.census.gov/programs-surveys/decennial/2020/
data/01-Redistricting_File--PL_94-171/
[StateXX]/2020.pl.sha256`

Any byte-level modification to the P.L. 94-171 file changes the SHA-256 hash with probability $1 - 2^{-256}$ (i.e., with certainty for any practical computation).

4. Attack Surface 2: TIGER/Line Shapefiles

4.1. The Data Format

TIGER/Line shapefiles define the geographic boundaries of each census tract as a sequence of latitude/longitude coordinate pairs. Two tracts are adjacent if their boundary polygons share a segment of positive length (the “shared edge” criterion used by BISECT). The adjacency

graph $G = (V, E)$ is constructed from these shapefiles: V is the set of census tracts, and $(u, v) \in E$ iff tracts u and v share a boundary segment.

4.2. The Attack

A hostile actor who can modify TIGER/Line shapefiles can rewire the adjacency graph without changing population counts. Specifically:

1. **Adjacency insertion:** Slightly modify the boundary coordinates of two non-adjacent tracts so that they appear to share a boundary segment. This adds an edge to the adjacency graph, potentially enabling a bisection cut that was previously infeasible.
2. **Adjacency deletion:** Remove boundary coordinates shared between two adjacent tracts, breaking their adjacency. This removes an edge, forcing the bisection to find alternative cuts.
3. **Boundary shift:** Move a boundary line between two tracts (one heavily Democratic, one heavily Republican) by a small amount, effectively moving some people from one tract to the other without changing the total population.

Attack 3 (boundary shift) is the most potent because it operates below the GEOID level: the adversary does not change which GEOID a tract has or what its total population is, only where its geographic boundary lies. If the algorithm uses centroid-based population assignment (which BISECT does not—it uses census-provided population counts per GEOID), this attack could effectively move people between tracts.

4.3. Why Bisect Is Resistant to Boundary Shifts

The critical design decision in BISECT is that population is taken from the P.L. 94-171 file keyed by GEOID, not derived from shapefile geometry. A shapefile boundary shift does not change the P.L. 94-171 GEOID record and therefore does not change the population count used by the algorithm.

The only effect of a shapefile boundary shift is to change the adjacency relationship between tracts. This is the adjacency insertion/deletion attack described above.

4.4. Effect Magnitude

Adding or removing individual adjacency edges in the census tract graph has limited partisan effect because:

- Congressional district boundaries are determined by sequences of hundreds to thousands of edges in the graph; single-edge changes affect only the local margin of a bisection cut.
- The ConvergenceSweep algorithm [Della-Libera, 2026a] optimises over many possible cuts; a single spurious edge is unlikely to become the preferred cut unless it is in a highly competitive region.
- TIGER/Line shapefiles use high-precision coordinate pairs (6 decimal places of latitude/longitude, approximately 10-meter precision). Inserting a false shared edge requires introducing coordinate pairs that are inconsistent with the surrounding polygon geometry; a geometric consistency check can detect such insertions.

We estimate the maximum effect of a plausible (undetected) shapefile manipulation at $\Delta D \leq 0.4$ seats nationally. A manipulation producing a larger effect would require altering dozens of adjacency relationships in competitive states, which is geometrically implausible (non-convex polygon corners, sub-meter boundary precision).

4.5. Detection

bisect computes the adjacency graph hash:

```
SHA256(adjacency_graph_state_XX_2020.json) -> manifest
```

The adjacency graph is a deterministic function of the TIGER/Line shapefile. Any modification to the shapefile that changes adjacency relationships changes the adjacency hash.

Detection probability: $\delta = 0.98$. The 2% residual corresponds to modifications that change shapefile coordinates but do not change adjacency relationships (e.g., interior smoothing of non-boundary coordinates), which are genuinely undetectable but also have zero effect on the algorithm’s output.

5. Attack Surface 3: ACS Demographic Tables

5.1. The Data Format

American Community Survey (ACS) 5-year tables provide demographic estimates for census tracts, including:

- B02001: Race alone
- B03002: Hispanic or Latino origin by race
- B05001: Nativity and citizenship status

BISECT uses ACS tables to compute the VRA boost weight w_{vra} for census tracts in states with significant minority populations. The boost weight increases the edge weight between tracts that are both in VRA-eligible minority communities, encouraging the bisection to keep such communities intact.

5.2. The Attack

An adversary could manipulate ACS minority population estimates to:

1. Artificially inflate minority percentages in tracts in non-competitive areas, spreading VRA boost weights broadly and diluting their effect in areas where minority community protection matters.
2. Artificially deflate minority percentages in tracts in competitive areas with high minority concentrations, weakening the VRA boost and allowing the bisection to divide minority communities.

5.3. Effect Magnitude

From U.2’s w_{vra} sweep [Della-Libera, 2026b], the partisan effect of VRA boost weight variation across its full range is $\Delta D \leq 0.2$ seats nationally. An ACS manipulation that effectively sets w_{vra} to zero for specific tracts would be at most as consequential as removing VRA boost entirely, which U.2 shows produces ≤ 0.2 seats of shift.

More precisely: ACS manipulation affects only the weight assigned to VRA-eligible edges. From U.2 Table 5, the total VRA boost effect ranges from $w_{\text{vra}} = 0.2$ (minimal) to $w_{\text{vra}} = 0.6$

(strong). ACS manipulation can simulate any value in this range. The maximum effect is therefore $\Delta D \leq 0.2$ seats.

Combining with the saturation argument, ACS manipulation contributes $\Delta D \leq 0.1$ seats to the combined attack, as listed in Table 1.

5.4. Detection

`bisect fetch` computes SHA-256 hashes for all ACS tables used. ACS tables are also available for independent download at:

<https://www2.census.gov/programs-surveys/acs/>

Detection probability: ≈ 1.00 .

6. The Three-Level Audit Chain

BISECT implements a three-level SHA-256 audit chain that provides layered protection against data manipulation.

6.1. Level 1: Census Bureau Download Hash

`bisect fetch -year 2020 -workers 8` downloads all required Census files and computes SHA-256 hashes immediately upon receipt. The hashes are stored in the manifest alongside the Census Bureau’s published checksums.

```
bisect fetch --year 2020 --state NC
[INFO] Downloading p194171_nc_2020.pl
[INFO] SHA256: a7f3bc91...
[INFO] Comparing against Census Bureau checksum: MATCH
[INFO] Downloading t1_2020_37_tract.zip
[INFO] SHA256: d4e82ca5...
[INFO] ACS B02001 5-year: 2016-2020
[INFO] SHA256: f1c0a9b7...
[INFO] Manifest written: nc_2020_fetch.manifest.json
```

A party wishing to independently verify the download can fetch the same files directly from the Census Bureau website and compare hashes.

6.2. Level 2: Adjacency Construction Hash

After downloading shapefiles, BISECT constructs the census tract adjacency graph. The graph construction is deterministic: given the same TIGER/Line shapefile, the same adjacency graph is always produced. The SHA-256 hash of the adjacency graph is stored in the manifest.

```
bisect build nc_2020 --year 2020
[INFO] Building adjacency graph for NC
[INFO] Tracts: 2,672 Edges: 7,841
[INFO] Adjacency SHA256: 8b2a7d43...
[INFO] Adjacency hash stored in manifest
```

An independent party can verify the adjacency hash by running the same adjacency construction algorithm on their independently downloaded shapefile. Any discrepancy indicates shapefile manipulation.

6.3. Level 3: Final Plan Hash

The completed district assignment (one GEOID to district mapping per state) is hashed and stored in the manifest.

```
bisect label-verify nc_2020 --year 2020
[INFO] Replaying computation from manifest
[INFO] Census download hash: MATCH
[INFO] Adjacency hash: MATCH
[INFO] Plan hash: MATCH
[INFO] Verification: PASS
```

The three-level chain provides end-to-end provenance: any modification at any level propagates into a hash mismatch at that level and all subsequent levels.

6.4. Chain Security

Theorem 1 (Audit Chain Security). *Under SHA-256 (FIPS PUB 180-4 [National Institute of Standards and Technology, 2015]), an adversary cannot produce a modified input file that has the same SHA-256 hash as the original file without performing at least 2^{80} hash operations*

(birthday bound for collision resistance). At 10^{15} hash operations per second on current hardware, this requires $> 3 \times 10^7$ years of computation.

This is not a formal proof of SHA-256 collision resistance (which remains an open problem) but reflects the current state of cryptographic knowledge. No practical collision attack on SHA-256 is known as of 2026.

6.5. Verification by Third Parties

The DISTRICTING INTEGRITY ACT requires the redistricting manifest to be filed publicly with the Secretary of State. Any party—opposing counsel, expert witnesses, journalists, academic researchers—can independently verify the manifest using:

```
bisect label-verify <label> --year <year> --manifest <path>
```

This converts the audit chain from a self-certification into a publicly verifiable claim.

7. Disclosed Limitation: TLS Certificate Validation

7.1. The Limitation

BISECT uses HTTPS for all Census Bureau downloads. TLS certificate validation is performed against the system trust store (the operating system’s certificate authority database). BISECT does *not* implement certificate pinning—it does not hard-code the specific TLS certificate or public key of the Census Bureau’s server.

This creates a potential attack surface for man-in-the-middle (MITM) interception:

1. An adversary who can add a certificate authority to the system trust store (requires administrator access to the redistricting machine) could intercept TLS connections to `www2.census.gov` and serve manipulated data.
2. The manipulated data would be downloaded with a valid TLS certificate (from the adversary’s CA) and would not trigger a TLS validation error.
3. The BISECT SHA-256 hash would record the hash of the *manipulated* file, not the original. This hash would not match the Census Bureau’s published checksum, but the adversary controls the comparison too (if they also control the manifest).

This attack chain requires the adversary to simultaneously: (a) compromise the system trust store; (b) intercept network traffic to the Census Bureau; (c) serve manipulated data that is plausible enough to pass visual inspection; (d) modify the comparison checksum source.

7.2. Risk Assessment

This attack requires infrastructure-level compromise of the redistricting authority’s computing environment—comparable to compromising the system running the redistricting algorithm entirely. An adversary with this level of access could also directly modify the algorithm binary, the parameter files, or the output plan, bypassing the data layer entirely.

In the context of the DISTRICTING INTEGRITY ACT’s deployment model (redistricting run by a certified neutral authority with audited computing infrastructure), this attack is not plausible without a broader security failure.

7.3. Recommended Mitigation

For litigation-grade deployments, we recommend supplementary out-of-band hash verification:

1. Download Census P.L. 94-171 files via two independent network paths (e.g., direct download and Census FTP mirror).
2. Compare SHA-256 hashes between the two copies.
3. Additionally compare against hashes published by independent redistricting data services (e.g., IPUMS NHGIS, Harvard Election Data Archive).

This multi-source verification eliminates the TLS limitation without requiring certificate pinning.

Future versions of BISECT may implement certificate pinning using the Census Bureau’s published TLSA DNS records (RFC 6698), which would eliminate the attack surface entirely. This is tracked as a known limitation in the BISECT security documentation.

8. Combined Effect Analysis

8.1. Individual Bounds

From the preceding sections:

- Population manipulation ($\pm 1\%$ per tract): $\Delta D \leq 0.4$ seats
- Shapefile adjacency manipulation: $\Delta D \leq 0.4$ seats
- ACS demographic manipulation: $\Delta D \leq 0.1$ seats

8.2. Why the Combined Effect Does Not Exceed 0.4 Seats

All three attacks share the same geometric bottleneck: competitive-district geography. A competitive district is one where the Democratic vote share is within 3–5 percentage points of 50%. In 2020, there were approximately 45 competitive congressional districts nationally.

Each of the three attacks can, at most, flip one or two of these districts. But flipping the same district via two different attacks does not double the effect—a district either flips or it doesn’t. The attacks are therefore largely substitutes, not complements.

Furthermore, the three attacks require different kinds of data access: population manipulation requires write access to P.L. 94-171 files; shapefile manipulation requires write access to TIGER/Line files; ACS manipulation requires write access to ACS tables. An adversary who has write access to all three simultaneously has effectively compromised the entire data pipeline—which is already detected by the audit chain.

8.3. The $\pm 1\%$ Threshold

We use $\pm 1\%$ as the attack magnitude because:

- It is within the Census Bureau’s acknowledged margin of error for tract-level population estimates (differential privacy noise at the block level, aggregated to tracts).
- It is small enough that a manipulation of this magnitude might be attributed to legitimate data correction rather than intentional gaming.
- It is precisely the threshold at which the effect is still within B.7’s noise floor.

For manipulations $> 1\%$ per tract, the effect grows approximately linearly until saturation at competitive-district thresholds. At $\pm 5\%$ per tract, ΔD might reach 1–2 seats nationally, but such a manipulation would be easily detectable through comparison with independently-sourced Census data.

8.4. Below the Noise Floor

B.7 [Della-Libera, 2026c] establishes that the seed-variance noise floor for BISECT is approximately $\sigma \approx 0.8$ seats nationally. The combined V2 attack maximum of 0.4 seats is within 0.5σ of zero— indistinguishable from natural algorithmic variation.

This means that even a successful (undetected) combined V2 attack would produce partisan outcomes that cannot be confidently attributed to the attack rather than to legitimate algorithmic variation. The attack is both bounded and non-attributable.

9. Conclusion

Input data manipulation is a genuine gaming vector for algorithmic redistricting: an adversary with access to the data pipeline can modify population counts, shapefile boundaries, or demographic tables before the algorithm processes them. However, the vector is both bounded and defended.

Bounded: even a combined attack across all three data sources produces at most $\Delta D \leq 0.4$ Democratic seats nationally, well within the B.7 seed-variance noise floor. The constraint is geometric—competitive-district geography cannot absorb more than a small fraction of the adversary’s manipulation before the districts are non-competitive anyway.

Defended: BISECT’s three-level SHA-256 audit chain (download hash, adjacency hash, plan hash) makes undetected manipulation computationally infeasible. Any modification that exceeds the $\pm 1\%$ noise threshold produces a hash mismatch detectable by any party who independently downloads and verifies the Census source files.

The disclosed TLS limitation (no certificate pinning) is a real but bounded risk. It requires infrastructure-level compromise of the redistricting computing environment, which is a broader security failure not specific to the data manipulation gaming vector. The recommended mitigation (multi-source hash verification) eliminates this risk for litigation-grade deployments.

The combined finding of R.1 and R.2 for the Track R program: neither parameter gaming

nor input data manipulation can produce a meaningful partisan outcome undetected. The remaining plan-affecting gaming vector (V3: geographic definition) is analysed in R.3; the audit chain formalization as a legal defense is developed in R.4.

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